

Cell-Type Annotation

in Single-Cell Genomics

A gentle introduction for biology and chemistry audiences

Single-Cell RNA-seq

Gene Expression Matrix

Marker Genes

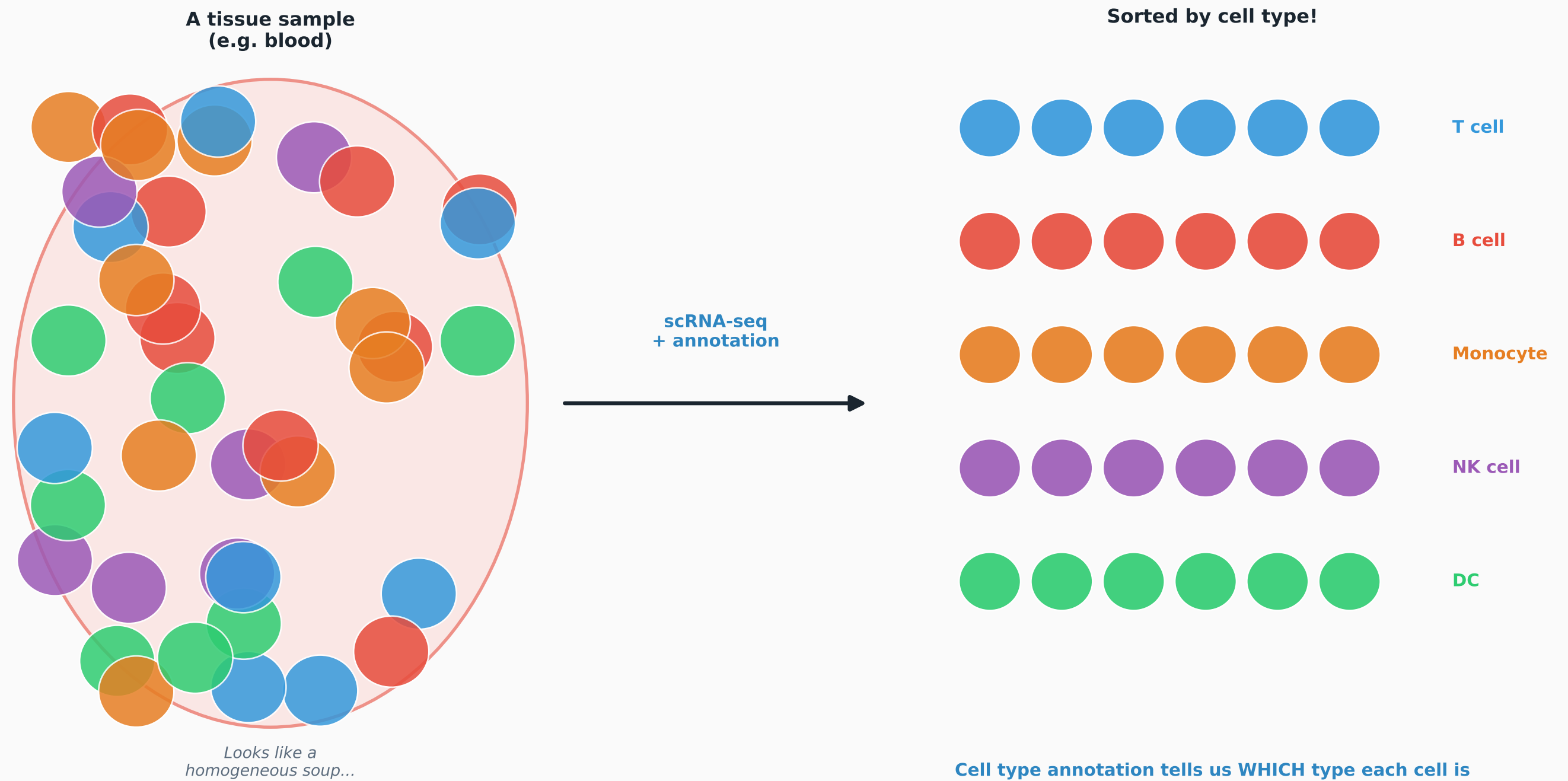
Zero-Shot Annotation

Reference Mapping

Scripps Research Hackathon 2026

The Central Question

A tissue contains many different cell types — but how do we tell them apart?



What is Gene Expression?

The molecular language that defines cell identity

The Central Dogma of Molecular Biology



Chemistry analogy:

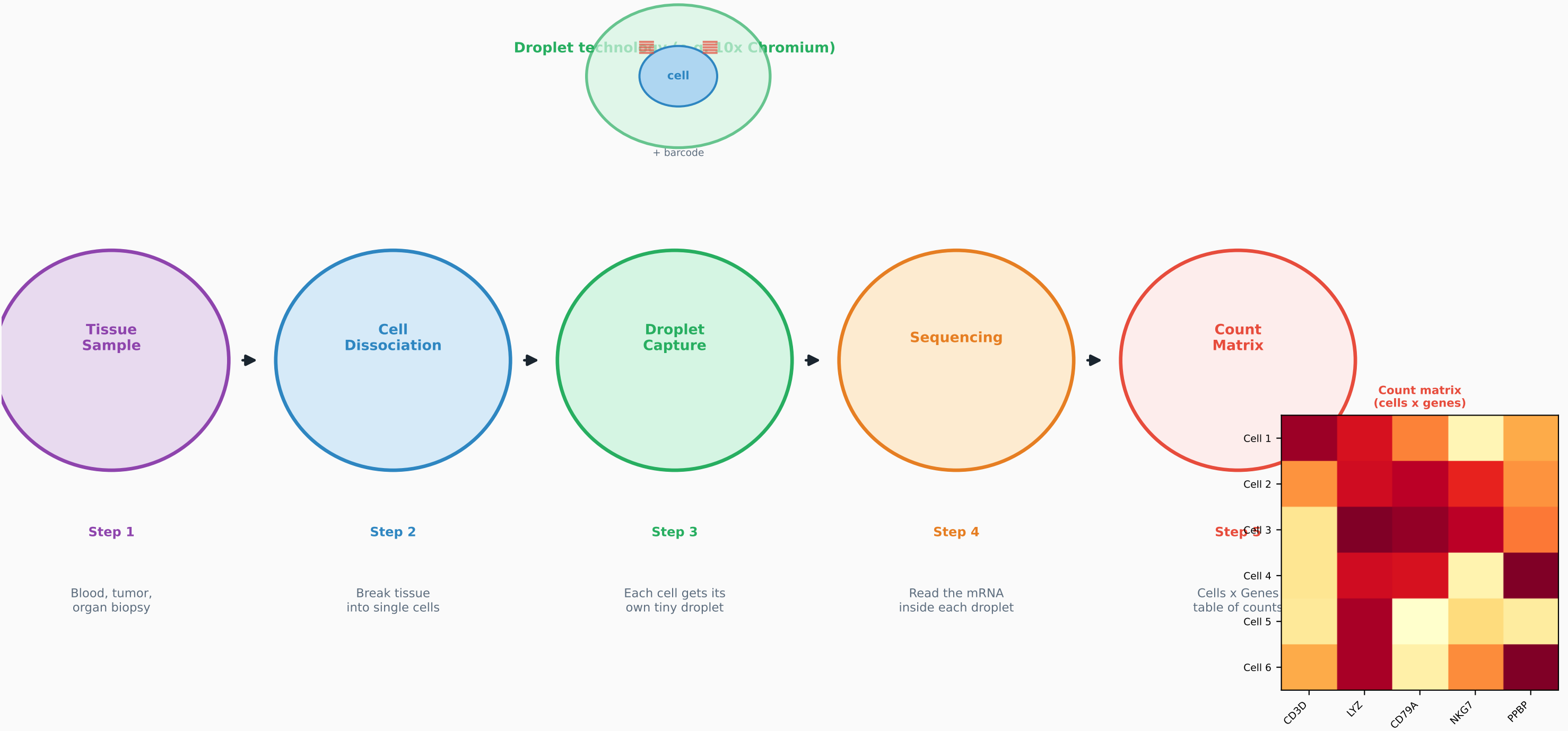
Gene expression is like a chromatogram — instead of measuring molecular masses, we count how many times each gene's mRNA is detected in a cell. High counts = gene is 'active' in that cell.

Key insight

Different cell types activate different sets of genes. This is what gives each cell type its unique identity.

Single-Cell RNA Sequencing (scRNA-seq)

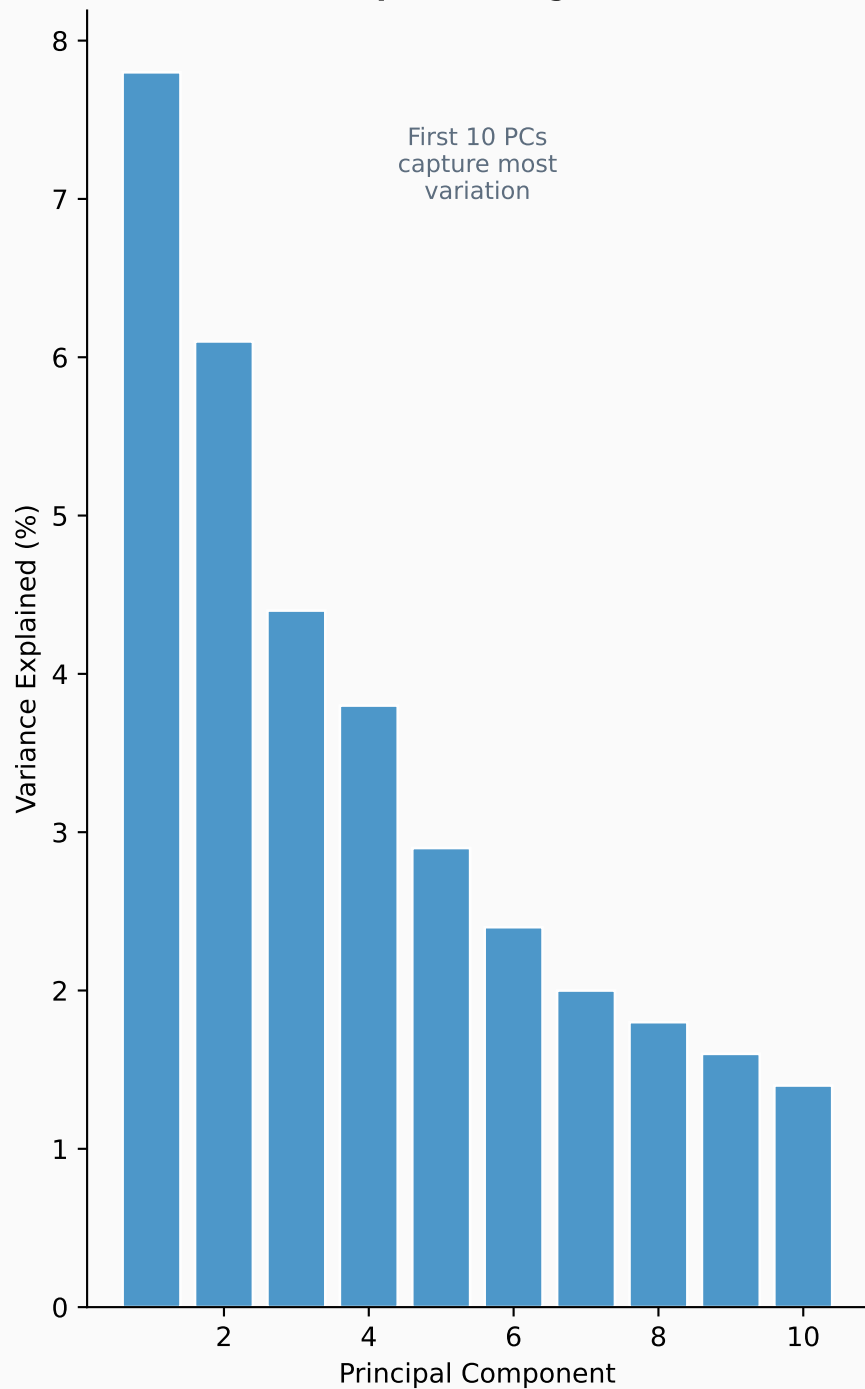
Measuring gene activity in thousands of individual cells simultaneously



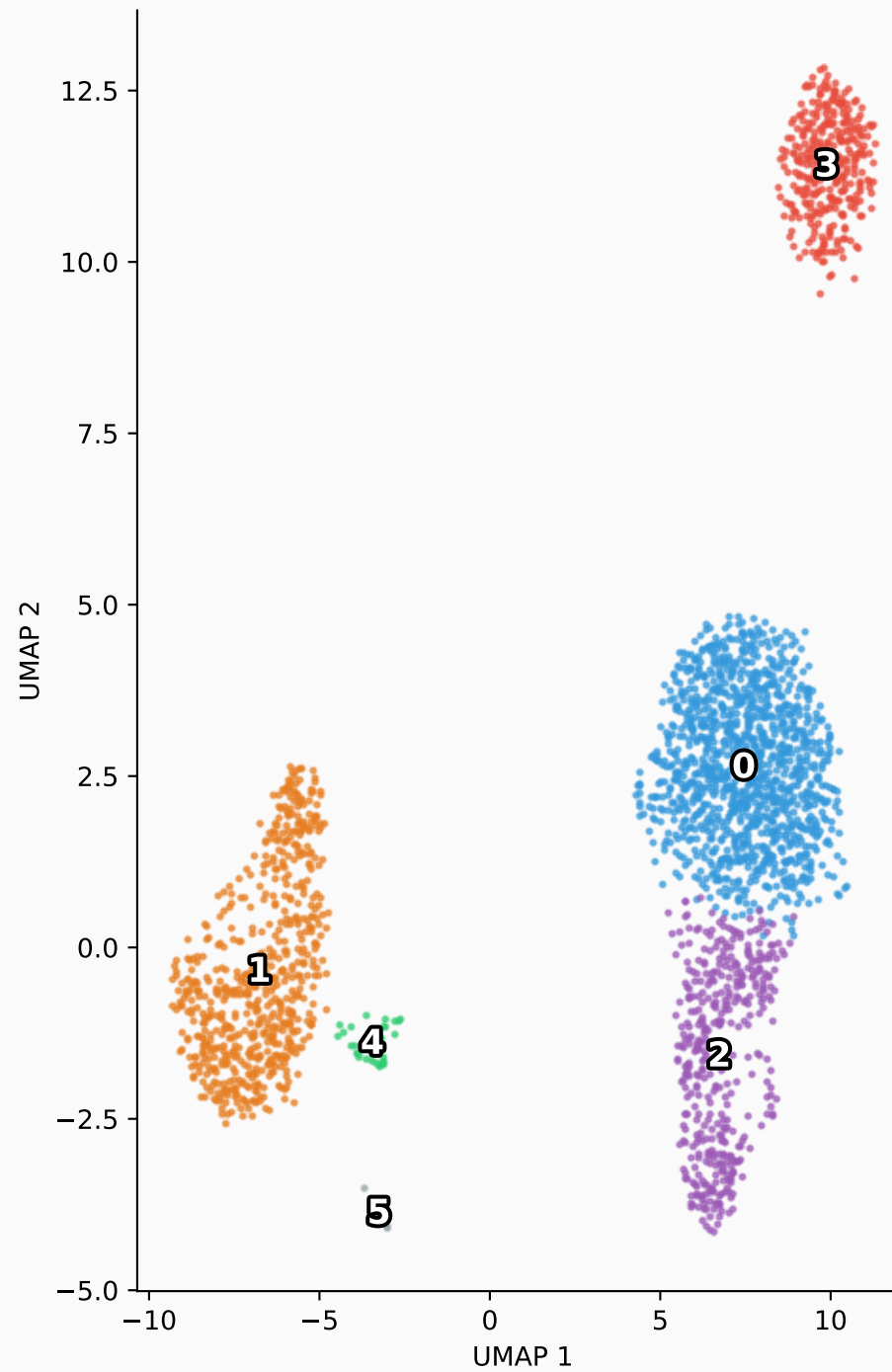
From Data Matrix to UMAP: Finding Cell Groups

Dimensionality reduction collapses 20,000 genes into a 2D map — similar cells cluster together

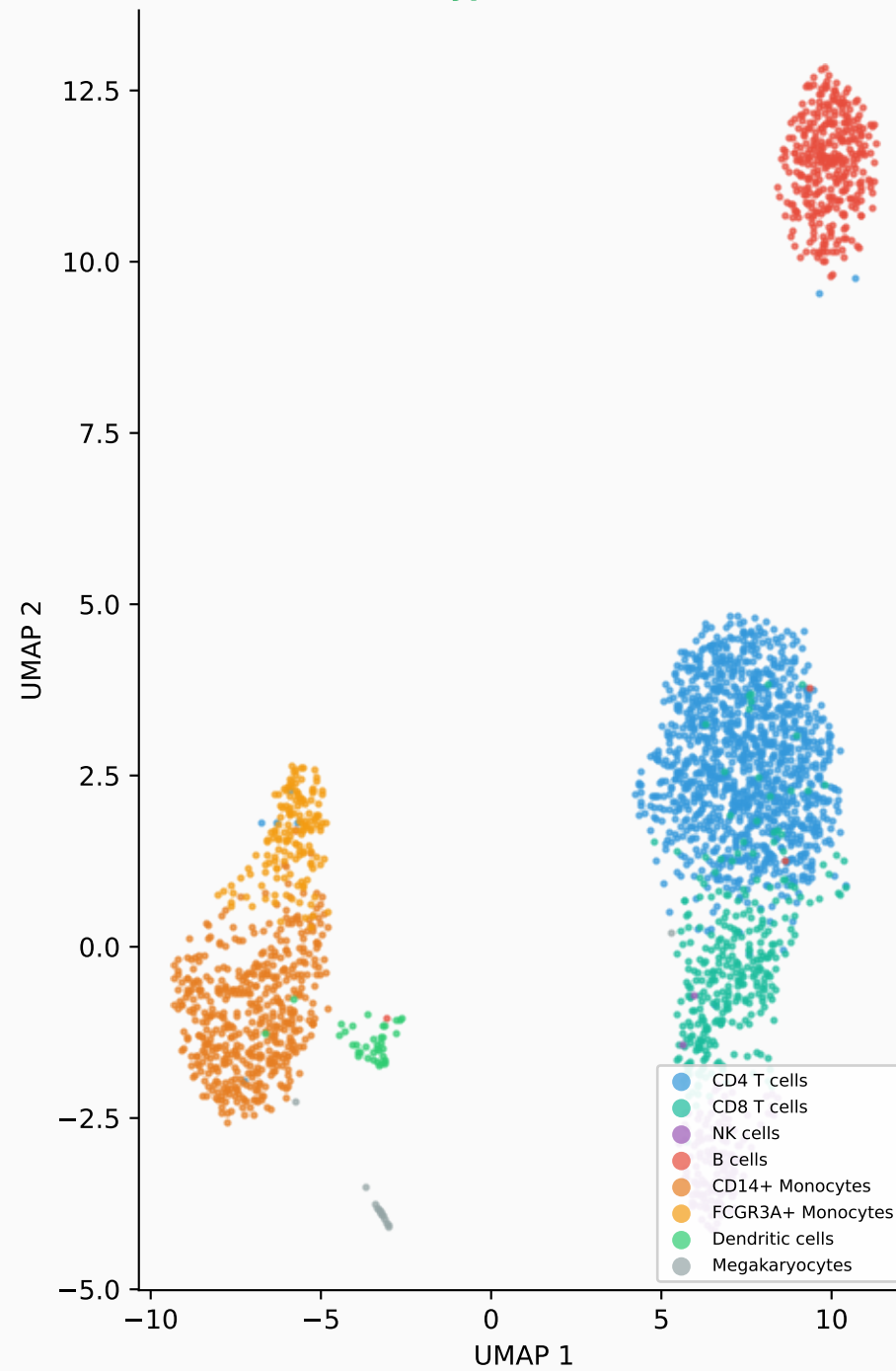
Step 1: PCA
(compress 20k genes)



Step 2: UMAP
(unlabelled clusters)



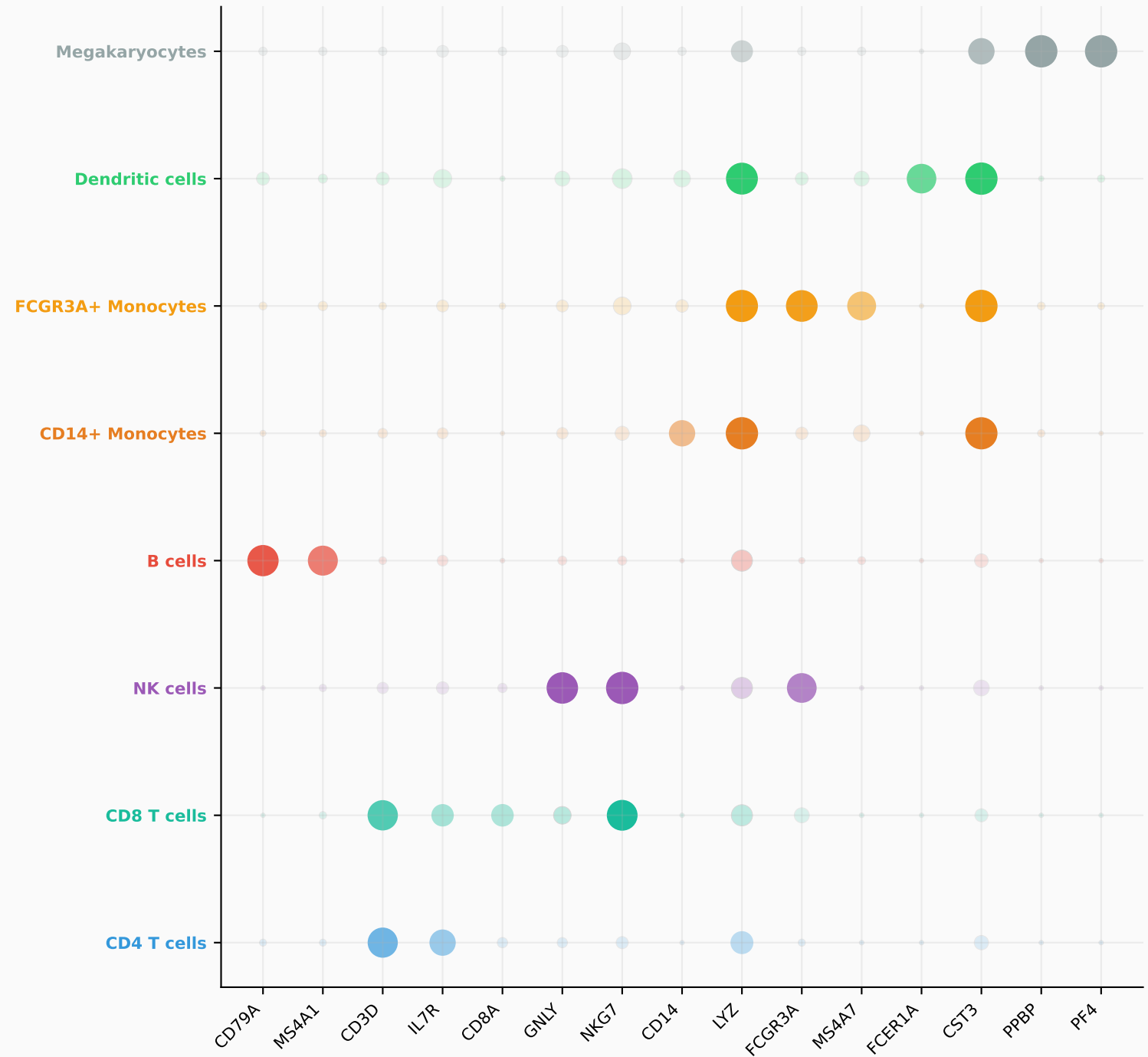
Step 3: Annotated!
(cell types revealed)



Marker Genes: The Molecular Fingerprints of Cell Identity

Some genes are uniquely and highly expressed in specific cell types — these are 'marker genes'

Dot Plot: Marker Gene Expression
(size = % cells expressing, colour = expression level)



Classic PBMC Marker Genes

Gene(s)	Cell Type	Function
 CD3D / CD3E	T cells	T-cell receptor complex
 CD79A / MS4A1	B cells	B-cell receptor complex
 CD14 / LYZ	CD14+ Monocytes	Innate immunity / phagocytosis
 FCGR3A	FCGR3A+ Monocytes	FC receptor (non-classical)
 GNLY / NKG7	NK cells	Cytotoxic granule proteins
 FCER1A	Dendritic cells	IgE receptor
 PPBP / PF4	Megakaryocytes	Platelet factors

High expression of a marker gene in a cluster = strong evidence for that cell type

Method 1: Zero-Shot Annotation with Marker Genes

Annotate without training data — use biological prior knowledge directly

"Zero-shot" = no training data needed

Works like a human expert looking up a reference book

UMAP with
Leiden clusters
(unlabelled)

Find top marker
genes per cluster
(rank_genes_groups)

Known marker
gene dictionary
OR LLM (Claude)

Cluster labelled
with cell type
name

Example: Cluster 3 has top genes CD79A, MS4A1, CD79B

Cluster 3
top genes:
CD79A, MS4A1

Known markers:
CD79A = B cells
MS4A1 = B cells

Annotation:
"B cells"
Confidence: HIGH

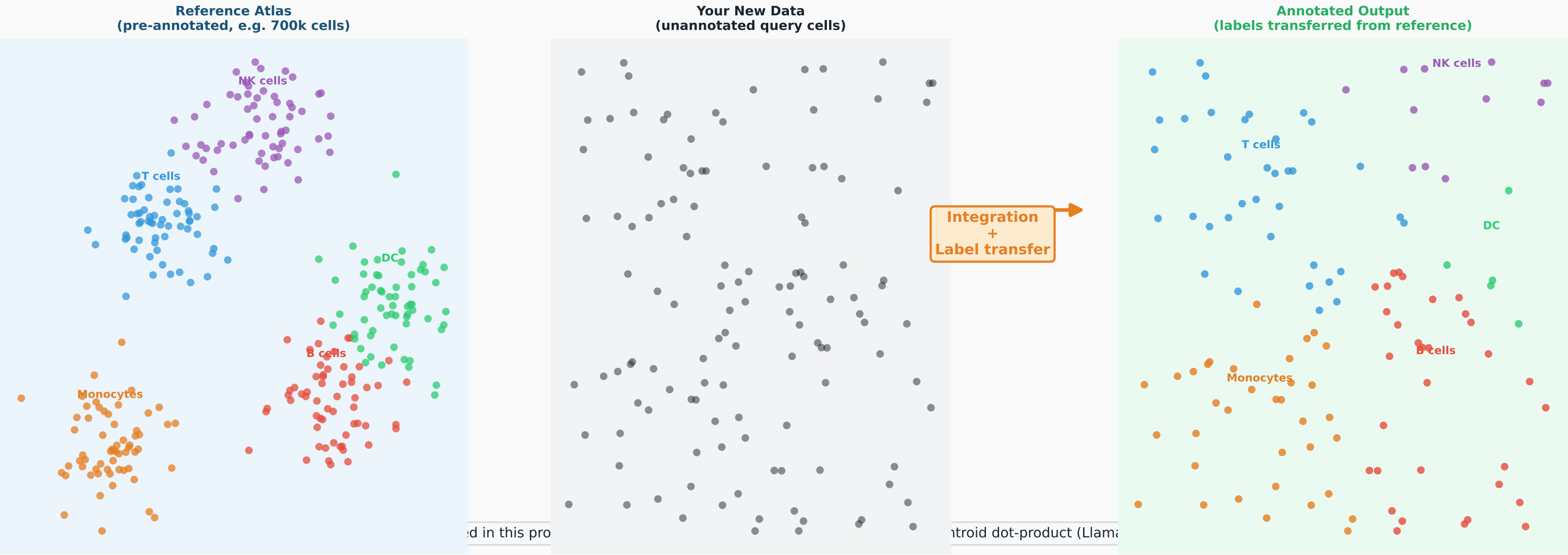
LLM variant:
Send marker genes
to Claude/Llama
Get label back

This project!

Method 2: Reference Mapping

Compare your cells to a large, already-annotated atlas and borrow their labels

Analogy: Reference mapping is like GPS navigation. Your new cells are an unknown location — the reference atlas is the map.



Why Does Cell-Type Annotation Matter?

Annotated cells unlock biological and clinical insights

Disease Research

Identify which cell types are expanded or depleted in cancer, autoimmune disease, or infection

Drug Target Discovery

Find the exact cell type where a drug acts — or should act — in the body

Immune Profiling

Track how T, B, NK cells change before and after vaccination or treatment

Cell Atlas Projects

Build comprehensive maps of every cell type in the human body (Human Cell Atlas)

Biomarker Discovery

Identify rare cell populations that predict patient outcomes or disease progression

Drug Development

Understand off-target effects: which cell types are affected by a new compound

Without cell-type annotation, single-cell data is just 20,000 numbers per cell — annotation gives it biological meaning.

Summary: Zero-Shot vs Reference Mapping

Two fundamentally different strategies for the same goal

	Zero-Shot (Marker Genes)	Reference Mapping
Core idea	Match cluster's top genes to known marker gene lists	Align cells to a pre-annotated reference atlas
Prior knowledge needed	Curated marker gene lists (literature / LLM knowledge)	A labelled reference dataset (e.g. Human Cell Atlas)
Granularity	Cluster-level label (all cells in a cluster get same label)	Cell-level label (each cell annotated independently)
Strengths	- No reference data required - Fast and interpretable - Works with any organism	- Cell-level resolution - Handles novel datasets well - Leverages large training sets
Weaknesses	- Limited by cluster resolution - Misses rare cell types - Requires good marker genes	- Needs a good reference - Batch effects can mislead - Reference may be incomplete
Accuracy (PBMC3k test)	~85% (Claude & Llama) 76% (Gemma)	~89% (Claude / CellTypist) 77% (Gemma / Scanorama) 73% (Llama / dot-product)
Best when...	Reference data unavailable or working with novel tissue	High accuracy needed and a good reference exists

Both methods are complementary — for best results, use both and compare!